

Summary Data on Acid Cropland and Lime Requirements in Sierra Leone

GAIA

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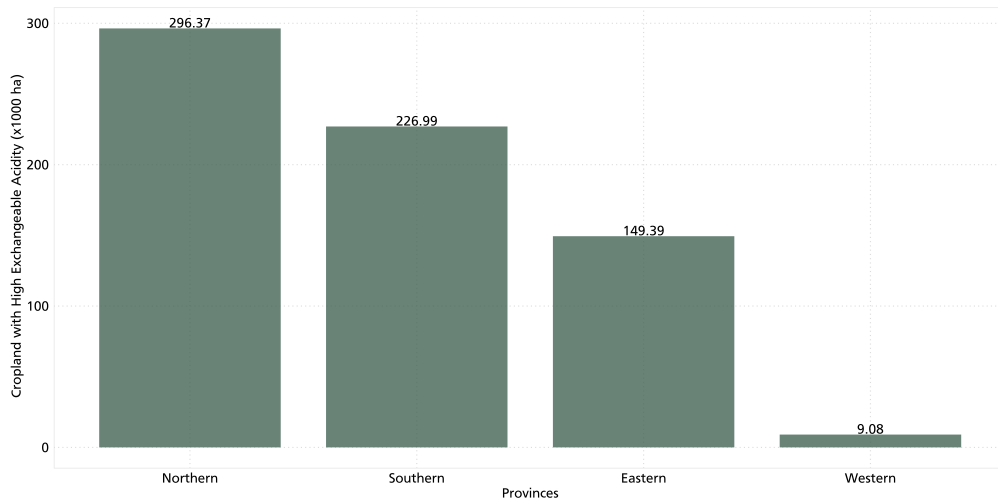
Outlook

- 1 Introduction
- 2 Acidic Cropland
- 3 Yield Loss
- 4 Lime Requirement
- 5 Yield Response
- 6 Additional Production
- 7 ROI for Lime Application
- 8 Profitability of Lime Application

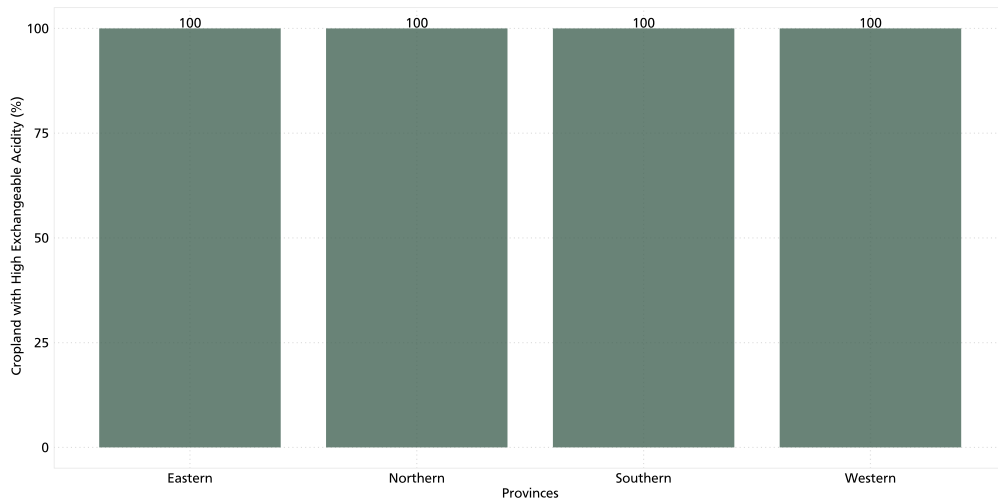
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Sierra Leone**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production, ROI and profitability from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

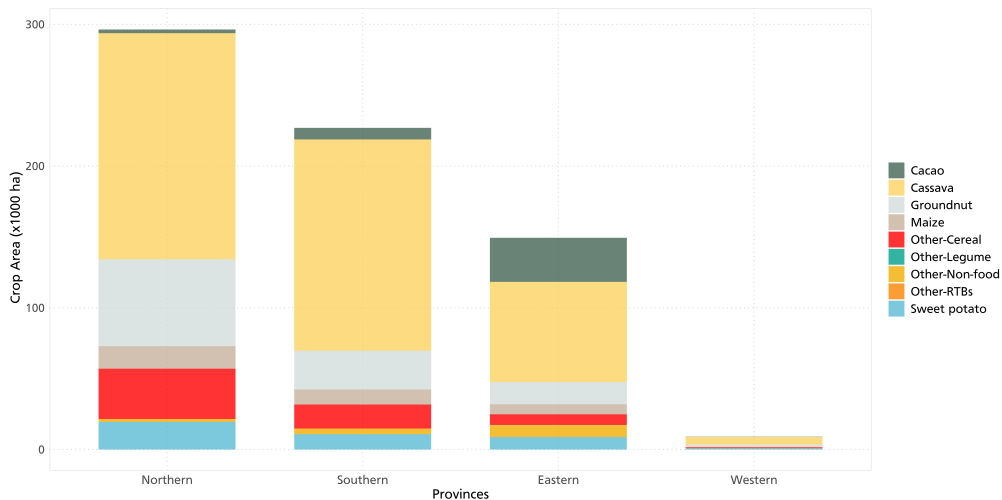
Top 10 Provinces in Sierra Leone with the highest acidic cropland



Top 10 Provinces in Sierra Leone with the highest acidic cropland (Percentage of total cropland)



Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland



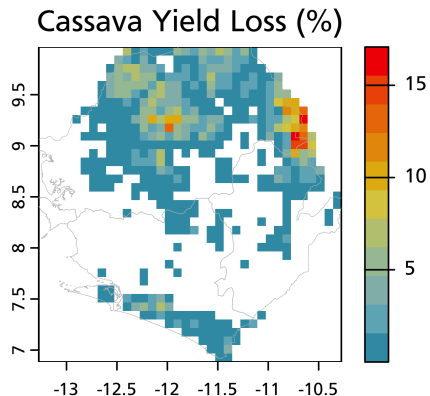
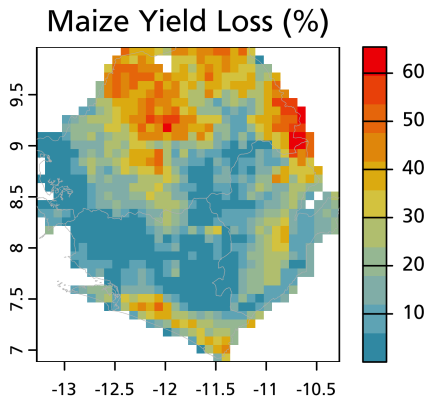
└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

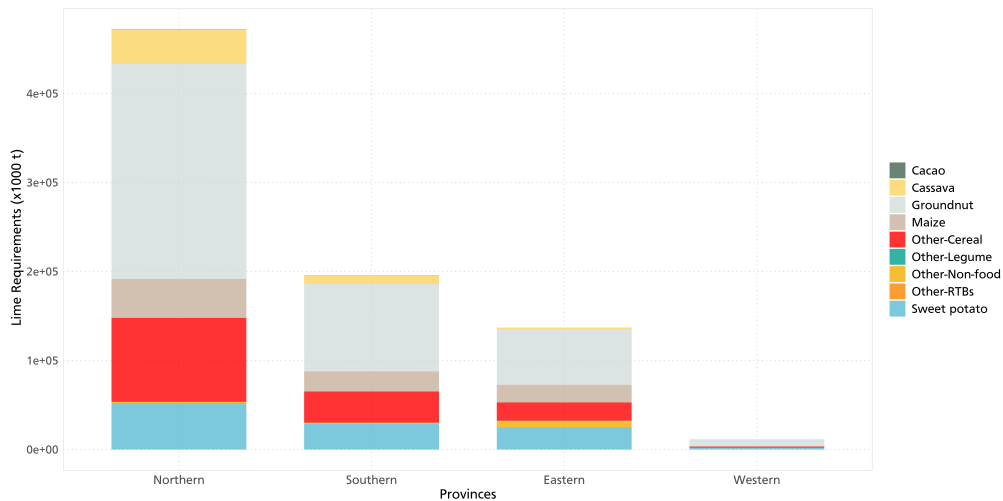
Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland (table)

Provinces	Cacao	Cassava	Groundnut	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sweet potato
Eastern	31.11	70.67	15.67	7.12	7.40	0.09	8.49	-	8.83
Northern	2.58	159.57	61.30	15.89	35.48	-	2.07	0.01	19.47
Southern	8.19	149.04	27.41	10.60	16.94	-	4.01	-	10.80
Western	0.21	5.40	1.54	0.35	0.64	-	0.08	-	0.86

Yield loss due to soil acidity for Maize and Cassava in Sierra Leone



Lime requirements (t) for acid soil remediation in Sierra Leone

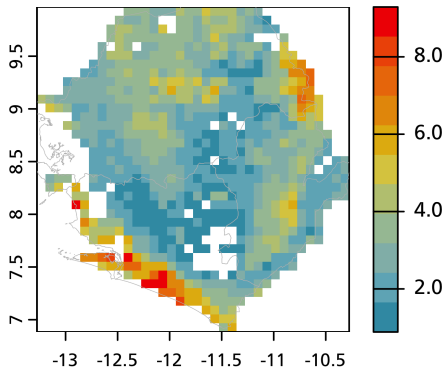


Lime requirements (1000 t) for acid soil remediation in Sierra Leone (table)

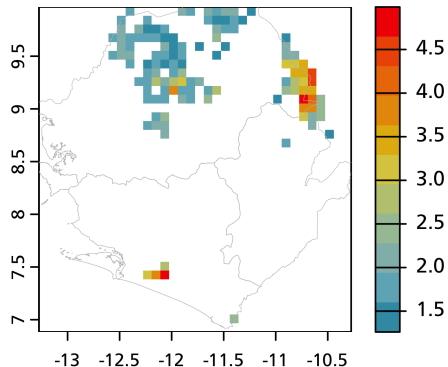
Provinces	Cacao	Cassava	Groundnut	Maize	Other-Cereal	Other-Legume	Other-Non-food	Sweet potato	Other-RTBs
Eastern	0.07	1.84	62.49	19.83	20.37	0.21	7.52	24.79	-
Northern	0.45	38.03	241.84	43.75	94.25	-	1.48	52.08	0.06
Southern	0.20	9.60	98.06	22.58	35.01	0.02	1.39	28.78	-
Western	-	-	7.71	0.67	1.07	-	-	2.11	-

Lime rate for Maize and Cassava in Sierra Leone

Maize Lime Rate (t/ha)

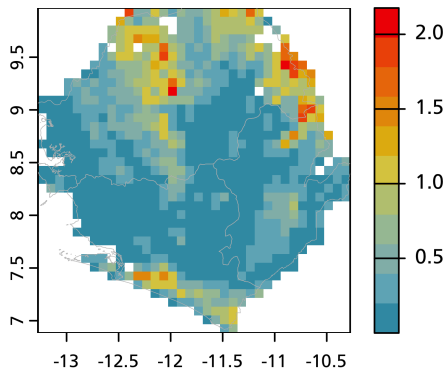


Cassava Lime Rate (t/ha)

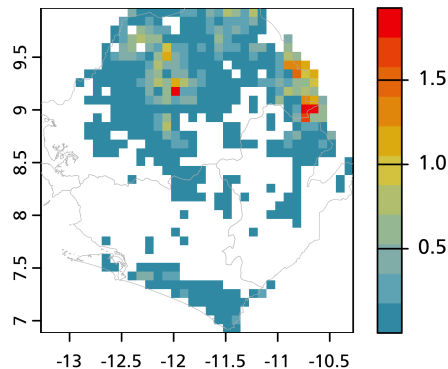


Yield response to lime application in Sierra Leone

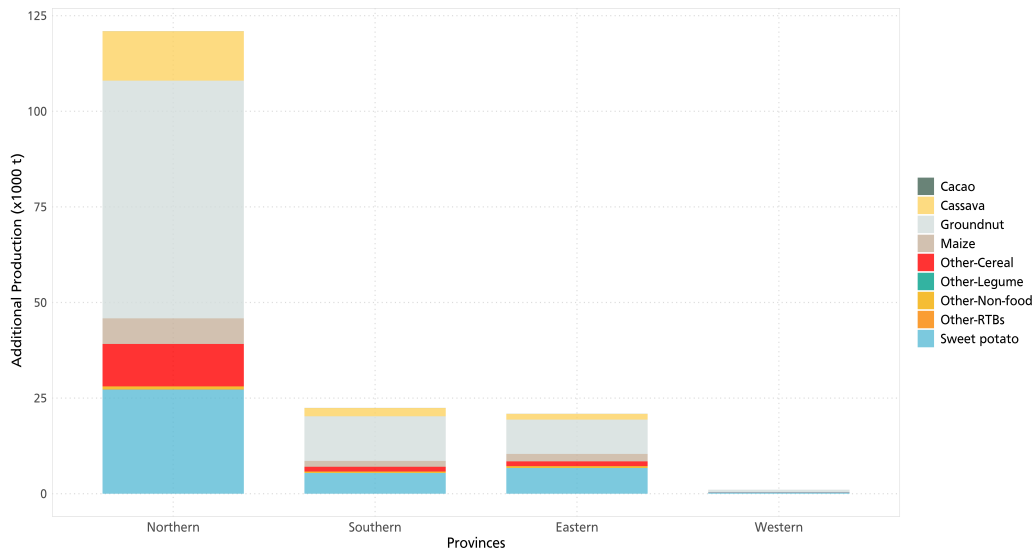
Maize Yield Response (t/ha)



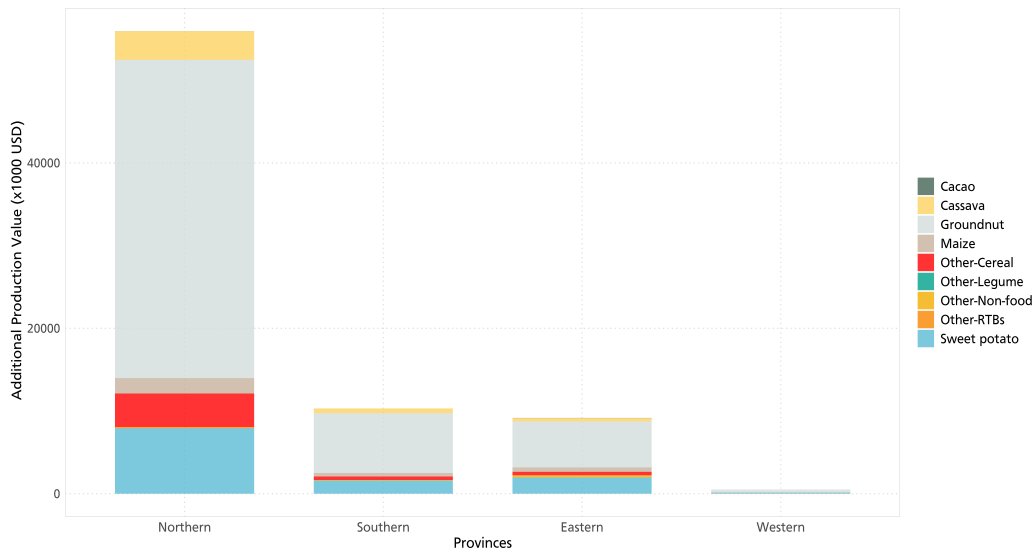
Cassava Yield Response (t/ha)



Potential for additional production (tone) in top 10 acidic Provinces in Sierra Leone

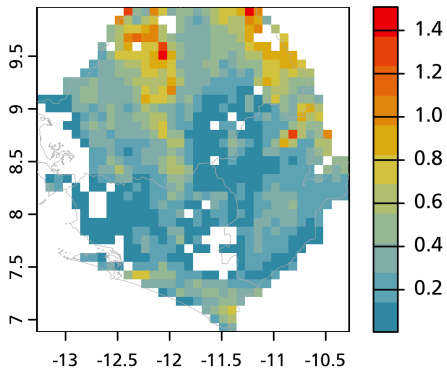


Potential for additional production value (US\$) in top 10 acidic Provinces in Sierra Leone

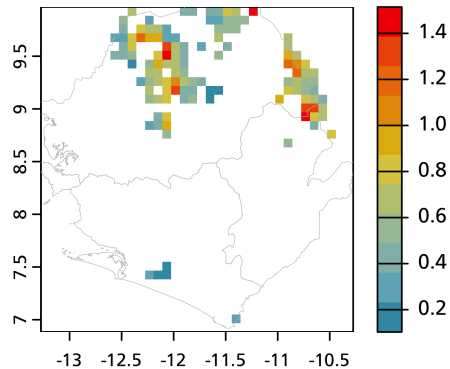


Return on investment for lime application in Sierra Leone

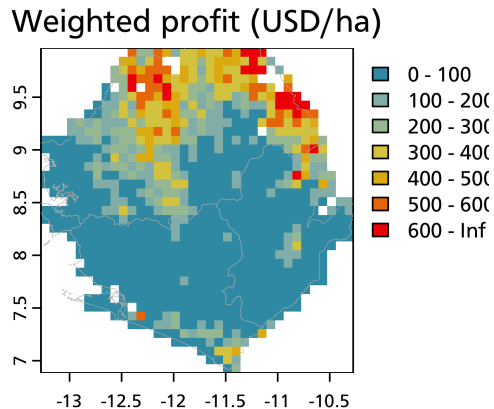
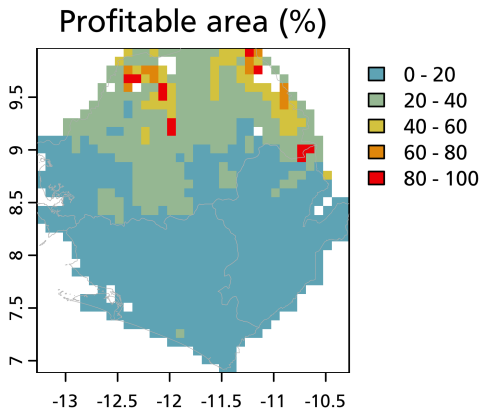
Maize ROI (USD/USD)



Cassava ROI (USD/USD)



Profitability of lime application in Sierra Leone



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.